

essential feature of CAD software's is their ability to seamlessly interpret geometric information and convert them to binary units. To understand the complexity of this process imagine this: what is easier, teaching someone using non-symbolic, non-verbal terms (things computers can't understand) the way to add two numbers or teaching someone using non-symbolic, non-verbal terms the shape of a gear?

The level of complexity in programming needed to comprehend the human mind's perception of three dimensional geometric shapes, compositions, details and designs is not as easily discovered as number theory. Since the mind has evolved to dynamically render its environments through the information from the eyes in three dimensions makes the transferring of that information to software extremely difficult.

This is where design software's make all the difference. By creating a new, efficient and constantly evolving language for computers to interpret how the three dimensional world exists is the first step towards making 3D printing a reality. The next step is allowing features that would allow users to alter and fabricate software based design's that would follow the rules of the real world. With the massive strides in processing power and graphics acceleration, we have arrived at a place where software's are able to distill three dimensional designs within their memory with greater accuracy and physics than most humans.

The method with which CAD software's are able to accomplish this task is by capturing the shape of any three dimensional object across the three coordinates - x, y and z. For example, a computer can now easily fabricate a model of a cube with data on its width, depth and height due to geometrical symmetry. But when we begin dealing with complex and irregular shapes, such as those found in nature - flowers, rocks, fractal patterns - the process becomes exponentially complex. These "general" physical shapes are created as sets of millions or billions of x-y-z coordinates that constitute simpler shapes.

The software is capable of not only storing this vast data sets but also interpolating their physical consequences. This ability allows users to simply deal with model representations of the code in the form of 3D models while the program works out the corresponding calculations. Each tweak and alteration in the design made is calculated, adjusted and corrected for physically accurate rendering across the x-y-z coordinates. Its important to emphasis that even seemingly basic shapes like a cylinder that are vectored across height and diameter are as complicated as a jet engine design due to